Steklov Maps and Hierarchical Poincaré–Steklov Solves

Promoted Jacobi Strong Residuals, Co-normal Stabilization,
Divergence-Form Elliptic Operators, and Arbitrary-Dimensional Stencils

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21 October 2026 — mathematical working draft

Abstract

We formulate the hierarchical Poincaré–Steklov (HPS) method from the continuous inhomogeneous Dirichlet problem through local Dirichlet-to-Neumann (DtN) decomposition, a promoted Jacobi strong-residual Petrov–Galerkin leaf construction, affine moment-space Steklov maps, two-subdomain merging, root boundary solves, and downward volume reconstruction. The leaf retains the existing expanded strong-form differential operator and stacked QR least-squares extension. Its augmented outward co-normal flux is the physical co-normal moment minus the positively weighted local trace–skeleton mismatch. The augmentation is the exact skeleton gradient of the reduced local least-squares energy and defines a symmetric positive semidefinite operator. We distinguish this algebraic fact from a still-conditional global coercivity statement in the presence of advection and reaction. Source-dependent affine terms and an explicit reusable upward source pass are included. The later treatment of arbitrary- dimensional Jacobi recurrences, natural stencils, differentiation, parameter promotion, and multiplication is retained from the prior note.

1 Continuous divergence-form Dirichlet and co-normal problem

Let $\Omega \subset \mathbb{R}^D$ be a Lipschitz domain partitioned into nondegenerate affine D-simplices and, recursively, unions of simplices. On a subdomain $K \subset \Omega$, consider

$$\mathcal{L}_K u = -\nabla \cdot (A_K \nabla u) + \boldsymbol{b}_K \cdot \nabla u + c_K u = f_K \quad \text{in } K, \qquad u = h_K \quad \text{on } \partial K. \tag{1}$$

The diffusion tensor $A = A^\top$ is uniformly positive definite within each coefficient region, and $A, \boldsymbol{b}, c$ have enough elementwise regularity for the traces below. Our chosen physical outward co-normal derivative is

$$q_K(u) = \boldsymbol{n}_K^\top A_K \nabla u \quad \text{on } \partial K. \tag{2}$$

Piecewise-discontinuous coefficients are permitted, with transmission conditions

$$[u] = 0 \quad \text{and} \quad q_A(u_A) + q_B(u_B) = 0 \tag{3}$$

for opposing outward normals in the absence of singular interface sources. This is the divergence-form *co-normal* problem. If $\boldsymbol{b} \cdot \boldsymbol{n}$ is also continuous across an interface, the advective contributions cancel in the total flux $\boldsymbol{b}u - A\nabla u$ as well. For a discontinuous normal advective field, the co-normal and total-transport transmission models should not be conflated.

Assume the Dirichlet realization of $\mathcal{L}_K$ is uniquely solvable for the data under consideration. For example, a standard sufficient condition for the corresponding energy argument is uniform ellipticity, suitable coefficient regularity, and $c_{\text{eff}} := c - \frac{1}{2}\nabla \cdot \boldsymbol{b} \geq 0$; other well-posed coefficient regimes are

possible. Denote the zero-boundary solution operator by G_K and the homogeneous interior Dirichlet extension by H_K:

$$\mathcal{L}_K(H_K h) = 0, \qquad H_K h|_{\partial K} = h, \tag{4}$$

$$\mathcal{L}_K(G_K f) = f, \qquad G_K f|_{\partial K} = 0. \tag{5}$$

Linearity and uniqueness give the exact decomposition

$$u = H_K h + G_K f. \tag{6}$$

Applying the outward co-normal trace to each component defines the *continuous Dirichlet-to-Neumann operator* N_K and the source-induced co-normal flux q_K^f:

$$N_K h := q_K(H_K h), \qquad q_K^f := q_K(G_K f), \qquad \boxed{q_K(u) = N_K h + q_K^f.} \tag{7}$$

In standard Sobolev terminology $N_K : H^{1/2}(\partial K) \to H^{-1/2}(\partial K)$, interpreted appropriately for the stated coefficient assumptions. The source-induced term is affine data, not part of N_K. If $\boldsymbol{b} \neq 0$, N_K is generally nonsymmetric.

What HPS propagates. For each subdomain K, hierarchical Poincaré–Steklov reduction propagates the pair (N_K, q_K^f): a homogeneous boundary response and a particular source response. Two child subdomains are joined by matching the Dirichlet trace and balancing their opposing outward co-normal fluxes. This is the continuum motivation for the discrete maps derived below.

1.1 Continuous two-subdomain transmission

Suppose $K = K_A \cup K_B$, with shared interface $I = \partial K_A \cap \partial K_B$ and external portions E_A, E_B. Prescribe h_A on E_A and h_B on E_B, and introduce the unknown common interface Dirichlet value γ:

$$h_{K_A} = (h_A, \gamma), \qquad h_{K_B} = (h_B, \gamma). \tag{8}$$

For continuous co-normal traces, the interface equation is

$$\Big[N_A(h_A, \gamma) + q_A^f\Big]_I + \Big[N_B(h_B, \gamma) + q_B^f\Big]_I = 0. \tag{9}$$

Once solved for γ, the external components of the two child fluxes define (N_K, q_K^f) on $E_A \cup E_B$. Repeated application constructs the root DtN map before any external boundary law is solved.

2 Jacobi strong Petrov–Galerkin residual and face skeleton

Let $K = F_K(\widehat{K})$, $x = x_{K,0} + B_K \widehat{x}$, with $\widehat{K}$ the reference D-simplex. Set

$$V_n(K) = \mathbb{P}_n^D(K), \qquad W_R(K) = \mathbb{P}_R^D(K), \qquad \Lambda_n(F) = \mathbb{P}_n^{D-1}(F), \tag{10}$$

with $M = \binom{n+D}{D}$ volume modes, $m_R = \binom{R+D}{D}$ strong-residual modes, and $k_F = \binom{n+D-1}{D-1}$ modes per face. The current full-degree elliptic implementation chooses $R = n$. We represent $u_h \in V_n(K)$ by the orthonormal Jacobi-simplex basis $\{\phi_j^\kappa\}$, and test the strong PDE residual against $\{\psi_i^{\kappa_{\mathrm{res}}}\}$, with

$$\kappa_{\mathrm{res}} = \kappa + \mathbf{2}, \qquad w_\kappa(x) = \lambda_0^{\kappa_D - \frac{1}{2}} \prod_{j=1}^{D} \lambda_j^{\kappa_{j-1} - \frac{1}{2}}, \qquad \kappa_j > -\tfrac{1}{2}. \tag{11}$$

The base parameter $\kappa_j = 1/2$ gives the unweighted physical polynomial family; general admissible κ is allowed. The canonical $(D-1)$-simplex face basis is fixed consistently on all faces, including their orientations.
For $u_h = \sum_j a_{K,j}\phi_j^\kappa$ define the strong Petrov–Galerkin moments

$$\mathcal{R}_{K,i}(u_h; f) = |\det B_K| \int_{\widehat{K}} [(\mathcal{L}_K u_h - f) \circ F_K]\, \psi_i^{\kappa_{\mathrm{res}}} w_{\kappa_{\mathrm{res}}}\, \mathrm{d}\widehat{x}. \tag{12}$$

Write these moments as

$$\boldsymbol{r}_K = L_K a_K - \boldsymbol{f}_K, \tag{13}$$

where the strong matrix L_K and the promoted source moments $\boldsymbol{f}_K$ use exactly the same residual family, geometry factors, and quadrature convention. In code paths that store the residual as $L_K a_K + f_{\mathrm{int},K}$, one has $f_{\mathrm{int},K} = -\boldsymbol{f}_K$ with this continuum sign convention.
The divergence-form adapter is permitted to construct L_K from the expanded strong coefficients

$$-A_{ij}\partial_{ij}u + \left(b_j - \sum_i \partial_i A_{ij}\right)\partial_j u + cu, \tag{14}$$

using the existing Jacobi differentiation, promotion, and multiplication machinery. No weak-form stiffness matrix is substituted.

Let $T_K \in \mathbb{R}^{n_{\partial K}\times M}$ be the stacked physical face trace map and $F_K \in \mathbb{R}^{n_{\partial K}\times M}$ the outward *physical co-normal moment* map. The common-skeleton vector $\lambda_K \in \mathbb{R}^{n_{\partial K}}$ and $T_K a_K$ use the same canonical face coordinate representation. The face space is orthonormal in the selected face weight (or must be paired with an explicit face Riesz matrix); we keep these normalizations visible when using physical unweighted boundary estimates.

2.1 Weighted residuals can be converted to physical norms

The promoted family does not prevent us from estimating the physical strong residual. Two orthonormal Jacobi bases of the *same* polynomial space $\mathbb{P}_R^D$ are related by an invertible finite connection/ladder matrix. Let

$$C_{0\leftarrow\mathrm{res}} : [p]_{\kappa_{\mathrm{res}}} \longmapsto [p]_{\kappa_0}, \qquad \kappa_0 = (1/2, \ldots, 1/2). \tag{15}$$

JPolyD's parameter-connection machinery (including the arbitrary-valid- parameter mode of the `KMat` builder) supplies such a map in principle; it need not be sparse. For a polynomial residual with promoted *coefficient* vector ρ, physical L^2 integration has an explicitly computable Gram matrix

$$\|p\|^2_{L^2(K)} = \rho^\top G_{0\leftarrow\mathrm{res},K}\rho, \quad G_{0\leftarrow\mathrm{res},K} = C_{0\leftarrow\mathrm{res}}^\top M_{0,K} C_{0\leftarrow\mathrm{res}}. \tag{16}$$

Here $M_{0,K}$ includes the known reference normalization and physical Jacobian. If $\boldsymbol{r}_K$ is stored as *moments* rather than coefficients, one first applies the residual-family Riesz/Gram inverse (and the known geometry scale) to obtain ρ. Consequently, the physical and promoted norms on $\mathbb{P}_R^D$ are equivalent, with constants given by the extreme eigenvalues of this computable positive matrix. This is an algebraic route to explicit finite-n bounds, not an assumed asymptotic norm equivalence. For nonpolynomial or insufficiently resolved variable coefficients, $L_K a_K - \boldsymbol{f}_K$ describes a *projected* strong residual. The connection map converts that projected polynomial exactly; it does not control the unrepresented residual tail or quadrature aliasing. Such terms need separate consistency bounds in a full convergence argument.

3 Discrete leaf: strong residual and stabilized trace extension

For fixed skeleton data λ_K and promoted source moments $\boldsymbol{f}_K$, define the present local stacked least-squares extension by

$$\boxed{a_K = \arg\min_a \mathcal{J}_K(a; \lambda_K, \boldsymbol{f}_K), \qquad \mathcal{J}_K = \tfrac{1}{2} s_K^2 \|L_K a - \boldsymbol{f}_K\|_2^2 + \tfrac{1}{2} \|T_K a - \lambda_K\|_{W_K}^2.} \tag{17}$$

The interior row scale $s_K > 0$ and positive definite face penalty $W_K = W_K^\top \succ 0$ are discretization parameters. The active implementation takes W_K diagonal in canonical orthonormal face coordinates and uses a dense QR factorization of the stacked matrix

$$\mathbb{A}_K = \begin{bmatrix} s_K L_K \\ W_K^{1/2} T_K \end{bmatrix}, \qquad {}_K = \begin{bmatrix} s_K \boldsymbol{f}_K \\ W_K^{1/2} \lambda_K \end{bmatrix}. \tag{18}$$

Define

$$H_K = s_K^2 L_K^\top L_K + T_K^\top W_K T_K. \tag{19}$$

The elementary well-posedness criterion is exact.

Proposition 1 (Unique local polynomial extension)**.** *For $s_K > 0$ and $W_K \succ 0$, the minimizer in* (17) *is unique if and only if* $\ker L_K \cap \ker T_K = \{0\}$.

Proof. For any a, $a^\top H_K a = s_K^2 \|L_K a\|_2^2 + \|T_K a\|_{W_K}^2$. It vanishes precisely on the stated kernel intersection. Therefore $H_K \succ 0$ exactly when the stacked system has full column rank. □

The normal equations yield an affine volume response

$$\boxed{a_K = U_{\lambda,K} \lambda_K + c_{f,K}, \qquad U_{\lambda,K} = H_K^{-1} T_K^\top W_K, \qquad c_{f,K} = U_{f,K} \boldsymbol{f}_K, \quad U_{f,K} = H_K^{-1} s_K^2 L_K^\top.} \tag{20}$$

These inverses denote linear solves; the dense QR path computes the multiple right-hand sides without explicitly forming H_K^{-1}. In particular, $U_{\lambda,K}$ and $U_{f,K}$ can be cached and reused for multiple source functions.

3.1 Reduced least-squares energy and its boundary gradient

Let $\mathcal{J}_{K,\mathrm{red}}(\lambda_K; \boldsymbol{f}_K)$ be (17) evaluated at its unique minimizer. Define

$$\boxed{P_K = W_K - W_K T_K U_{\lambda,K} = W_K - W_K T_K H_K^{-1} T_K^\top W_K.} \tag{21}$$

Proposition 2 (Positive reduced residual boundary operator)**.** *The matrix P_K is symmetric positive semidefinite. For zero source,*

$$\lambda_K^\top P_K \lambda_K = \min_a \left(s_K^2 \|L_K a\|_2^2 + \|T_K a - \lambda_K\|_{W_K}^2 \right) \geq 0, \tag{22}$$

and

$$\ker P_K = T_K(\ker L_K). \tag{23}$$

For arbitrary source, the exact reduced gradient is

$$\boxed{\nabla_{\lambda_K} \mathcal{J}_{K,\mathrm{red}} = -W_K(T_K a_K - \lambda_K) = P_K \lambda_K - W_K T_K c_{f,K}.} \tag{24}$$

Proof. The normal equations give $H_K a_K = T_K^\top W_K \lambda_K + s_K^2 L_K^\top \boldsymbol{f}_K$. Inserting $a_K = U_{\lambda,K}\lambda_K$ into the homogeneous quadratic objective gives (22). The explicit matrix form is symmetric since H_K and W_K are symmetric. Positivity follows from the minimum of two squares. Equality in the homogeneous minimum is equivalent to an a with $L_K a = 0$ and $T_K a = \lambda_K$, which establishes the kernel identity. Finally, differentiating the objective with respect to fixed λ_K at its stationary minimizer eliminates the $\partial a_K/\partial\lambda_K$ term and gives (24). □

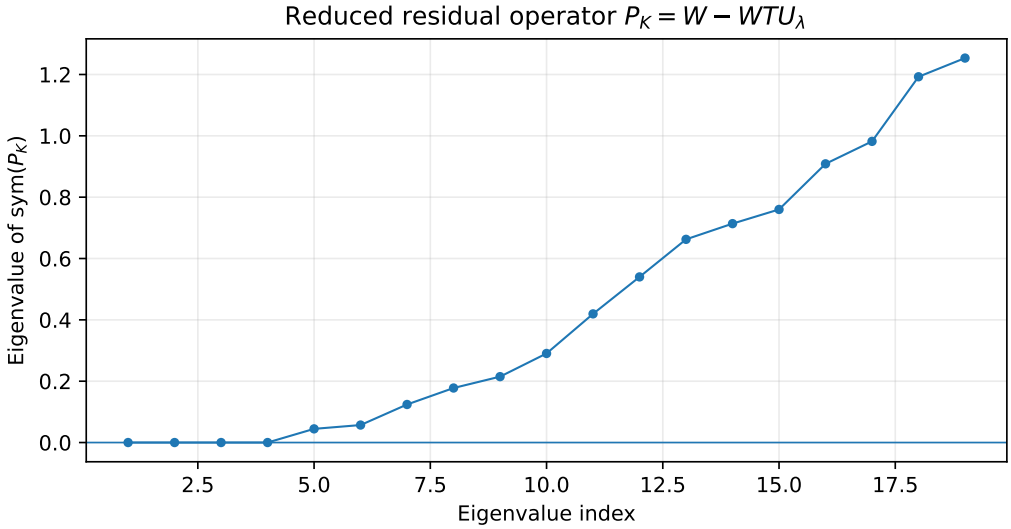


Figure 1: Numerical algebraic check of $P_K \succeq 0$. The bundled `verify_pk.py` tests $P_K - P_K^\top$, the reduced-energy identity, the affine gradient identity, and its eigenvalues. The displayed example is a *synthetic full-rank leaf*, not a claimed measurement from a JPolyD mesh. An optional NPZ argument accepts exported L_K, T_K, W_K, s_K from a real leaf.

3.2 Corrected augmented outward co-normal map

Write the mismatch as $\delta_K = T_K a_K - \lambda_K$. The numerical outward co-normal *moment* vector is defined by

$$\boxed{\mu_K = F_K a_K - W_K(T_K a_K - \lambda_K) = F_K a_K + \nabla_{\lambda_K}\mathcal{J}_{K,\mathrm{red}}.} \tag{25}$$

The physical co-normal map F_K uses the original divergence tensor A_K and the outward normal, not the expanded strong operator's first-order coefficient. The augmentation is thus exactly the skeleton derivative of the minimized residual energy. This fact justifies its sign but does *not* assert that the entire flux $F_K a_K$ is itself a gradient of a scalar functional.
Inserting (20) yields the complete affine leaf map

$$\boxed{\begin{aligned} \mu_K &= S_K\lambda_K + b_K, \\ S_K &= (F_K - W_K T_K)U_{\lambda,K} + W_K = F_K U_{\lambda,K} + P_K, \\ b_K &= (F_K - W_K T_K)c_{f,K} = (F_K - W_K T_K)U_{f,K}\boldsymbol{f}_K. \end{aligned}} \tag{26}$$

Here S_K is the moment-space *discrete Steklov/DtN approximation* corresponding to the continuous N_K, and b_K is the discrete source-induced co-normal response corresponding to q_K^f. They are not

identically equal to their continuum analogues at finite polynomial order. The discrete map acts on the complete per-element polynomial skeleton, including skeleton traces that do not extend to exact homogeneous polynomial solutions. P_K controls precisely that incompatibility.
For zero source, $S_K = F_K U_{\lambda,K} + P_K$. In particular, with $\mathrm{sym}(\cdot)$ the symmetric part,

$$\mathrm{sym}(S_K) = \mathrm{sym}(F_K U_{\lambda,K}) + P_K. \tag{27}$$

The PSD conclusion applies to P_K, not automatically to S_K. For nonsymmetric advection operators, $F_K U_{\lambda,K}$ can be nonsymmetric and its symmetric part need not be nonnegative. Global stability and invertibility of merge interface matrices require additional arguments or hypotheses.

3.3 A conditional PDE energy route, including reaction and advection

The continuous Green identity for a local polynomial u_h is

$$\langle q_K(u_h), u_h\rangle_{\partial K} = \|A_K^{1/2}\nabla u_h\|^2_{L^2(K)} + (c_{\mathrm{eff},K}u_h, u_h)_K + \frac{1}{2}\left\langle \boldsymbol{b}_K\cdot\boldsymbol{n}_K, u_h^2\right\rangle_{\partial K} - (\mathcal{L}_K u_h, u_h)_K, \tag{28}$$

where $c_{\mathrm{eff},K} = c_K - \frac{1}{2}\nabla\cdot\boldsymbol{b}_K$. For $\boldsymbol{f}_K = 0$, the exact least-squares normal equations give

$$-\lambda_K^\top W_K \delta_K = s_K^2\|L_K a_K\|_2^2 + \|\delta_K\|^2_{W_K}. \tag{29}$$

Combining this with $\lambda_K = T_K a_K - \delta_K$ and the physical Green identity gives, subject to proper face Riesz conversion,

$$\begin{aligned}\langle \mu_K, \lambda_K\rangle_{\partial K} = {} & \|A^{1/2}\nabla u_h\|^2_{L^2(K)} + (c_{\mathrm{eff}}u_h, u_h)_K + \tfrac{1}{2}\left\langle \boldsymbol{b}\cdot\boldsymbol{n}, u_h^2\right\rangle_{\partial K} \\ & + s_K^2\|L_K a_K\|_2^2 + \|\delta_K\|^2_{W_K} - \langle q_K(u_h), \delta_K\rangle_{\partial K} - (\mathcal{L}u_h, u_h)_K.\end{aligned} \tag{30}$$

The last two terms are not sign-definite; the first is estimated using face trace/inverse inequalities and a diffusion-scaled W_K, and the second using the physical-L^2 conversion in Section 2.1, plus estimates for any unresolved residual tail. For sufficiently positive effective reaction the bulk term is helpful; no additional *physical* reaction flux occurs on faces. Advection enters the boundary energy and may motivate a directional face weighting if the baseline penalty is insufficient, but is not postulated as mandatory here. Equation (30) is a *proof strategy*, not a claimed unconditional coercivity theorem. It also makes no hidden assumption that the entire HPS flux comes from a global least-squares Euler–Lagrange principle.

4 Exact algebraic HPS merge of two discrete subdomains

Consider two already-built (possibly composite) child HPS nodes A and B, with boundary data and augmented outward co-normal maps

$$\mu_A = S_A\lambda_A + b_A, \qquad \mu_B = S_B\lambda_B + b_B. \tag{31}$$

The skeleton is single-valued across the shared interface. Denote by E_A and E_B the child faces exterior to their parent, and by I their common set of interface faces. In the canonical orientation and ordering, write

$$\lambda_A = \begin{bmatrix}\lambda_{A,E}\\ \gamma\end{bmatrix}, \qquad \lambda_B = \begin{bmatrix}\lambda_{B,E}\\ \gamma\end{bmatrix}. \tag{32}$$

Physical normals on the interface are opposite, but the interface basis coefficients γ have been mapped to the *same* canonical face basis before merging. This is precisely what allows fluxes to add with their outward signs.
Partition the child maps into exterior and interface blocks:

$$\begin{bmatrix}\mu_{A,E}\\ \mu_{A,I}\end{bmatrix} = \begin{bmatrix}S_{A,EE} & S_{A,EI}\\ S_{A,IE} & S_{A,II}\end{bmatrix}\begin{bmatrix}\lambda_{A,E}\\ \gamma\end{bmatrix} + \begin{bmatrix}b_{A,E}\\ b_{A,I}\end{bmatrix}, \tag{33}$$

$$\begin{bmatrix}\mu_{B,E}\\ \mu_{B,I}\end{bmatrix} = \begin{bmatrix}S_{B,EE} & S_{B,EI}\\ S_{B,IE} & S_{B,II}\end{bmatrix}\begin{bmatrix}\lambda_{B,E}\\ \gamma\end{bmatrix} + \begin{bmatrix}b_{B,E}\\ b_{B,I}\end{bmatrix}. \tag{34}$$

Only the interface flux moments are eliminated. The parent exterior trace and outward co-normal numerical flux are

$$\lambda_P = \begin{bmatrix}\lambda_{A,E}\\ \lambda_{B,E}\end{bmatrix}, \qquad \mu_P = \begin{bmatrix}\mu_{A,E}\\ \mu_{B,E}\end{bmatrix}. \tag{35}$$

4.1 Interface continuity and the reconstruction map

The HPS interface balance is

$$\mu_{A,I} + \mu_{B,I} = 0. \tag{36}$$

Inserting the interface rows of (33)– (34) gives

$$\underbrace{(S_{A,II} + S_{B,II})}_{H_I}\gamma = -S_{A,IE}\lambda_{A,E} - S_{B,IE}\lambda_{B,E} - b_{A,I} - b_{B,I}. \tag{37}$$

Assume H_I is invertible. Define

$$\boxed{\begin{aligned} R_A &= -H_I^{-1}S_{A,IE},\\ R_B &= -H_I^{-1}S_{B,IE},\\ r &= -H_I^{-1}(b_{A,I} + b_{B,I}). \end{aligned}} \tag{38}$$

Then the downward reconstruction equation for the interior interface skeleton is

$$\boxed{\gamma = R_A\lambda_{A,E} + R_B\lambda_{B,E} + r.} \tag{39}$$

This is an algebraic Schur elimination and does not require the operator to be self-adjoint. For advective elliptic problems the matrix H_I may be nonsymmetric; invertibility and conditioning are properties to check, not consequences of $P_K \succeq 0$ alone.

4.2 Parent homogeneous and source-induced maps

Substitute (39) into the *external* rows of (33)–(34):

$$\begin{bmatrix}\mu_{A,E}\\ \mu_{B,E}\end{bmatrix} = \begin{bmatrix}S_{A,EE} & 0\\ 0 & S_{B,EE}\end{bmatrix}\begin{bmatrix}\lambda_{A,E}\\ \lambda_{B,E}\end{bmatrix} + \begin{bmatrix}S_{A,EI}\\ S_{B,EI}\end{bmatrix}(R_A\lambda_{A,E} + R_B\lambda_{B,E} + r) + \begin{bmatrix}b_{A,E}\\ b_{B,E}\end{bmatrix}. \tag{40}$$

Therefore the parent again has exactly the same affine Steklov form:

$$\boxed{\mu_P = S_P\lambda_P + b_P,} \tag{41}$$

where

$$\boxed{S_P = \begin{bmatrix} S_{A,EE} + S_{A,EI}R_A & S_{A,EI}R_B \\ S_{B,EI}R_A & S_{B,EE} + S_{B,EI}R_B \end{bmatrix},} \tag{42}$$

and

$$\boxed{b_P = \begin{bmatrix} b_{A,E} + S_{A,EI}r \\ b_{B,E} + S_{B,EI}r \end{bmatrix}.} \tag{43}$$

This is the full HPS merge, including the affine source contribution. The parent has the same data type as either child: a face-only operator S_P, a source-dependent shift b_P, and the data needed to reconstruct γ. Recursively, two triangles, tetrahedra, or arbitrary simplex subdomains build the root map using precisely the same formula.

4.3 Reusable source-only upward pass

The expensive matrices S_A, S_B, H_I, R_A, R_B depend on the operator and geometry, not on a particular source vector. Define

$$R_f := -H_I^{-1}, \qquad P_A := S_{A,EI}R_f, \qquad P_B := S_{B,EI}R_f. \tag{44}$$

For a new source, once child shifts are known, update

$$r = R_f(b_{A,I} + b_{B,I}), \tag{45}$$

$$b_P = \begin{bmatrix} b_{A,E} + P_A(b_{A,I} + b_{B,I}) \\ b_{B,E} + P_B(b_{A,I} + b_{B,I}) \end{bmatrix}. \tag{46}$$

This upward pass reuses the stored operators without repeating the matrix factorization or rebuilding the homogeneous DtN maps. The implementation stores R_f, P_A, P_B for this reason. At each leaf, a new volume source only updates $c_{f,K} = U_{f,K}\boldsymbol{f}_K$ and $b_K = (F_K - W_K T_K)c_{f,K}$.

5 Root Robin solve and downward recovery

At the root, the hierarchy has produced

$$\mu_{\text{root}} = S_{\text{root}}\lambda_{\text{root}} + b_{\text{root}}. \tag{47}$$

Consider a continuous boundary law

$$\alpha(x)u + \beta(x)q(u) = g(x) \qquad \text{on } \partial\Omega. \tag{48}$$

In the canonical root face moment representation, let M_α, M_β denote the appropriate face multiplication and projection operators. Imposing the Robin law on the numerical co-normal flux gives

$$\boxed{(M_\alpha + M_\beta S_{\text{root}})\lambda_{\text{root}} = \boldsymbol{g} - M_\beta b_{\text{root}}.} \tag{49}$$

For constant facewise coefficients these are scalar blocks. For variable coefficients, consistent integration of βq may require a higher-degree co-normal product correction when multiplication and face projection do not commute (see `high_order_conormal_robin_hps.tex` for dense materialization of the multiplication operator and high order correction details). The abstract root formula

(49) is understood to include that consistent moment operation. Because μ is augmented, a numerical Robin law generally differs from pointwise enforcement of the raw physical co-normal flux at finite n; the discrepancy is a validation diagnostic, not an algebraic contradiction.
Dirichlet conditions correspond to $\beta = 0$ and determine the trace; mixed/Robin conditions use the corresponding rows. In the pure Neumann Laplace limit, constants lie in the kernel; compatibility and a gauge constraint are necessary. For general advection/reaction operators, compatibility must be derived from that operator and must not be replaced mechanically by the Laplace formula (see `pure_neumann_jpolyd_hps.tex` for details).

Downward skeleton reconstruction. After solving the root boundary system, traverse the merge tree from root to leaves. For every parent split, partition its recovered exterior trace as $(\lambda_{A,E}, \lambda_{B,E})$, evaluate

$$\gamma = R_A \lambda_{A,E} + R_B \lambda_{B,E} + r, \tag{50}$$

and assemble each child's complete trace $\lambda_A = (\lambda_{A,E}, \gamma)$, $\lambda_B = (\lambda_{B,E}, \gamma)$. Continue recursively until all leaf skeleton traces are available.

Downward volume reconstruction. At each leaf, recover the Jacobi modal coefficients from the cached affine response:

$$\boxed{a_K = U_{\lambda,K} \lambda_K + c_{f,K}.} \tag{51}$$

The physical solution is then evaluated using the library's Jacobi basis evaluators, and physical gradients with their affine Jacobian maps. If desired, $\mu_K = S_K \lambda_K + b_K$ and the separately reconstructed raw co-normal moments $F_K a_K$ can be compared to monitor the trace-mismatch flux correction.

Complete computational ordering. For a fixed operator: (i) construct leaf U_λ, U_f, S and source-response maps; (ii) merge upward, storing interface inverses and reconstruction maps; (iii) for each new source, compute leaf c_f, b and propagate b and r upward; (iv) solve root Robin data; (v) reconstruct all internal traces downward; and (vi) reconstruct volume modal coefficients. The outward augmented co-normal penalty appears only in the leaf maps; all later HPS levels manipulate their affine face operators.

6 Algebraic verification and limits of the stability statement

The bundled `verify_pk.py` creates a full-rank illustrative leaf (or accepts exported L, T, W, s in an NPZ file), constructs

$$H = s^2 L^\top L + T^\top W T, \quad U_\lambda = H^{-1} T^\top W, \quad P = W - W T U_\lambda, \tag{52}$$

and checks: (a) $P = P^\top$ to roundoff; (b) the eigenvalues of $\mathrm{sym}(P)$ are nonnegative up to roundoff; (c) $\lambda^\top P \lambda$ equals the minimized homogeneous residual energy; and (d) the source-dependent gradient identity (24). These are exact algebraic identities under the rank assumption and do not depend on spatial dimension, Jacobi weights, or PDE coefficients. The companion script `verify_merge.py` independently checks the two-child interface balance and reconstructed parent affine map using synthetic nonsymmetric child blocks, including nonzero source shifts.
For the *complete* discrete Steklov map, positivity of P does not by itself prove coercivity of $S = FU_\lambda + P$, uniqueness of all merge interface solves, preservation of a maximum principle, or uniform

high-Péclet convergence. A full theorem additionally requires estimates for the physical co-normal response and its trace mismatch, the conversion of the promoted projected residual to a physical norm (Section 2.1), and any unrepresented coefficient/quadrature residual. Positive $c - \frac{1}{2}\nabla \cdot \boldsymbol{b}$ contributes helpful bulk control; reaction is not an independent physical face flux. An inflow-weighted penalty may be explored later if the baseline method fails outside the tested parameter range. None of these open analytical questions requires replacing the strong-form Jacobi operator or the HPS merge algebra.

7 Arbitrary-dimensional Jacobi operator recurrences and stenciling (retained from the original note)

Block sparsity assumptions on Jacobi simplex operators

Let

$$\Pi_n^D = \bigoplus_{j=0}^{n} \mathcal{H}_j^D$$

denote the decomposition of total-degree polynomials into homogeneous subspaces $\mathcal{H}_j^D$ of degree exactly j. We fix, for each j, a Jacobi simplex basis $\{\phi_\alpha^{(\kappa)} : |\alpha| = j\}$ of $\mathcal{H}_j^D$, ordered by a graded rule. We assume the existence of a canonical bijection between multi-indices α with $|\alpha| = j$ and local indices in $\mathcal{H}_j^D$.
All operators below are represented in this basis.

(i) Degree-lowering operators (DMat). Let

$$\mathcal{D} : \Pi_n^D \to \Pi_{n-1}^D$$

be a linear operator such that

$$\mathcal{D}\,\mathcal{H}_j^D \subset \mathcal{H}_{j-1}^D, \qquad j \geq 1.$$

Then the matrix of $\mathcal{D}$ has block structure

$$\mathcal{D} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ D_1 & 0 & 0 & \cdots & 0 \\ 0 & D_2 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & D_n & 0 \end{pmatrix},$$

where $D_j : \mathcal{H}_j^D \to \mathcal{H}_{j-1}^D$.
Sparsity assumption. There exists a finite set

$$\Delta_{\mathcal{D}} \subset \mathbb{Z}^D, \qquad \sum_{d=1}^{D} \Delta_d = -1,$$

independent of j, such that

$$\mathcal{D}\phi_\alpha^{(\kappa)} \in \operatorname{span}\big\{\phi_{\alpha+\Delta}^{(\kappa')} : \Delta \in \Delta_{\mathcal{D}}\big\}.$$

(ii) Degree-filtering operators (KMat). Let

$$\mathcal{K} : \Pi_n^D \to \Pi_n^D$$

be a linear operator arising from a *natural promotion* of Jacobi parameters,

$$\kappa_{\mathrm{tgt}} = \kappa_{\mathrm{src}} + e_r, \qquad r \in \{0, \ldots, D\},$$

corresponding to multiplication of the orthogonality weight by a single barycentric factor. Although $\mathcal{K}$ preserves the total polynomial degree globally, it is *not* degree-preserving at the level of homogeneous subspaces.

More precisely, $\mathcal{K}$ satisfies the block filtering property

$$\mathcal{K}\,\mathcal{H}_j^D \subset \mathcal{H}_j^D \,\oplus\, \mathcal{H}_{j-1}^D, \qquad j \geq 0,$$

with the convention $\mathcal{H}_{-1}^D = \{0\}$. Equivalently, the matrix of $\mathcal{K}$ in the graded basis

$$\Pi_n^D = \bigoplus_{j=0}^{n} \mathcal{H}_j^D$$

has a two-band block structure:

$$\mathcal{K} = \begin{pmatrix} A_0 & B_0 & 0 & 0 & \cdots \\ 0 & A_1 & B_1 & 0 & \cdots \\ 0 & 0 & A_2 & B_2 & \cdots \\ \vdots & & \ddots & \ddots & \ddots \end{pmatrix},$$

where

$$A_j : \mathcal{H}_j^D \to \mathcal{H}_j^D, \qquad B_j : \mathcal{H}_{j+1}^D \to \mathcal{H}_j^D.$$

That is, writing u as a graded decomposition

$$u = \sum_{j=0}^{n} u_j, \quad u_j \in \mathcal{H}_j^{\kappa_{\mathrm{src}}},$$

we define the operator

$$\mathcal{K} : \Pi_n^{\kappa_{\mathrm{src}}} \to \Pi_n^{\kappa_{\mathrm{tgt}}},$$

so that

$$\mathcal{K}u = \sum_{i=0}^{n} (\mathcal{K}u)_i, \quad (\mathcal{K}u)_i \in \mathcal{H}_i^{\mathrm{tgt}}.$$

So, target block i recieves contributions from source blocks i and $i+1$. Now, if u only has one homogeneous component, $u \equiv u_j \in \mathcal{H}_j^{\mathrm{src}}$, then the only terms remaining in the sum over i from the formula

$$(\mathcal{K}u)_i = A_i u_i + B_i u_{i+1}$$

are $i = j$ via $A_j u_j$ and $i = j-1$ via $B_{j-1} u_j$. Then, we see that

$$\mathcal{K}u_j = (\mathcal{K}u)_j + (\mathcal{K}u)_{j-1} \Rightarrow \mathcal{K}\mathcal{H}_j \subset \mathcal{H}_j \oplus \mathcal{H}_{j-1}.$$

Sparsity assumption. There exist finite sets of multi-index offsets

$$\Delta_{\mathcal{K}}^{(0)},\ \Delta_{\mathcal{K}}^{(-1)} \subset \mathbb{Z}^D,$$

independent of j, such that for every basis function $\phi_\alpha^{(\kappa_{\mathrm{src}})}$ with $|\alpha| = j$,

$$\mathcal{K}\phi_\alpha^{(\kappa_{\mathrm{src}})} \in \mathrm{span}\Big(\{\phi_{\alpha+\Delta}^{(\kappa_{\mathrm{tgt}})} : \Delta \in \Delta_{\mathcal{K}}^{(0)}\} \,\cup\, \{\phi_{\alpha+\Delta}^{(\kappa_{\mathrm{tgt}})} : \Delta \in \Delta_{\mathcal{K}}^{(-1)},\ |\alpha+\Delta| = j-1\}\Big).$$

The sets $\Delta_{\mathcal{K}}^{(0)}$ and $\Delta_{\mathcal{K}}^{(-1)}$ encode, respectively, the intra-degree couplings (A_j blocks) and the one-degree-down couplings (B_j blocks).

Stencil discovery. To determine these coupling patterns, we assemble a dense reference operator $\mathcal{K}^{(n_{\text{test}})}$ at a small test degree n_{test} and restrict it to the representative row block

$$\mathcal{K}_{\text{rep}}^{(n_{\text{test}})} = \mathcal{K}^{(n_{\text{test}})}\big|_{\mathcal{H}_{j_{\text{rep}}}^D \times \left(\mathcal{H}_{j_{\text{rep}}}^D \oplus \mathcal{H}_{j_{\text{rep}}+1}^D\right)}, \qquad j_{\text{rep}} = n_{\text{test}} - 1.$$

For each nonzero entry in this restricted block, we record the multi-index difference

$$\Delta = \alpha_{\text{row}} - \alpha_{\text{col}},$$

partitioning the resulting offsets according to whether they correspond to $\mathcal{H}_j \leftarrow \mathcal{H}_j$ or $\mathcal{H}_j \leftarrow \mathcal{H}_{j+1}$ couplings. Once these offset sets stabilize as n_{test} increases, they are reused to assemble $\mathcal{K}$ at arbitrary polynomial degree n without further dense computation.

(iii) Three-term recurrence operators (JMat). Let

$$\mathcal{J} : \Pi_n^D \to \Pi_{n+1}^D$$

be a linear operator satisfying

$$\mathcal{J}\,\mathcal{H}_j^D \subset \mathcal{H}_{j-1}^D \oplus \mathcal{H}_j^D \oplus \mathcal{H}_{j+1}^D.$$

Then the matrix of $\mathcal{J}$ has block tridiagonal structure:

$$\mathcal{J} = \begin{pmatrix} B_0 & C_0 & 0 & \cdots & 0 \\ A_1 & B_1 & C_1 & \ddots & \vdots \\ 0 & A_2 & B_2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & C_{n-1} \\ 0 & \cdots & 0 & A_n & B_n \end{pmatrix},$$

with

$$A_j : \mathcal{H}_j^D \to \mathcal{H}_{j-1}^D, \quad B_j : \mathcal{H}_j^D \to \mathcal{H}_j^D, \quad C_j : \mathcal{H}_j^D \to \mathcal{H}_{j+1}^D.$$

Sparsity assumption. There exists a finite set

$$\Delta_{\mathcal{J}} \subset \mathbb{Z}^D, \qquad \sum_{d=1}^{D} \Delta_d \in \{-1, 0, +1\},$$

independent of j, such that

$$\mathcal{J}\phi_\alpha^{(\kappa)} \in \operatorname{span}\{\phi_{\alpha+\Delta}^{(\kappa')} : \Delta \in \Delta_{\mathcal{J}}\}.$$

Delta-based representation of operator couplings

Let $\mathcal{O}$ be a linear operator acting on the graded polynomial space

$$\Pi_n^D = \bigoplus_{j=0}^{n} \mathcal{H}_j^D,$$

where $\mathcal{H}_j^D$ denotes the subspace of homogeneous polynomials of total degree j in D variables. Fix a Jacobi simplex basis $\{\phi_\alpha^{(\kappa)} : |\alpha| = j\}$ for each $\mathcal{H}_j^D$, indexed by multi-indices $\alpha \in \mathbb{N}^D$ with $|\alpha| = j$.

Coupling types. We assume that the action of $\mathcal{O}$ on a basis function $\phi_\alpha^{(\kappa)}$ can be written in the form

$$\mathcal{O}\phi_\alpha^{(\kappa)} = \sum_{\Delta \in \Delta_\mathcal{O}} c_\Delta(\alpha)\, \phi_{\alpha+\Delta}^{(\kappa')},$$

where:

- $\Delta_\mathcal{O} \subset \mathbb{Z}^D$ is a finite set of *multi-index shifts* ("deltas"),
- the coefficients $c_\Delta(\alpha)$ may depend on α and the Jacobi parameters, but
- the set $\Delta_\mathcal{O}$ itself is independent of the polynomial degree j.

Each $\Delta \in \Delta_\mathcal{O}$ represents a distinct *coupling type* between a source basis function indexed by α and a target basis function indexed by $\alpha + \Delta$.

Admissibility. For a fixed source multi-index α, a shift Δ is said to be *admissible* if the target multi-index

$$\beta = \alpha + \Delta$$

satisfies:

1. componentwise nonnegativity, $\beta_i \geq 0$ for all i, and
2. the appropriate degree constraint
$$|\beta| = |\alpha| + s,$$
where s is the degree shift associated with $\mathcal{O}$.

Only admissible shifts contribute to the action of the operator at a given α.

Stability of the delta set. For small degrees j, some coupling types may be absent due to boundary effects in the index set $\{\alpha : |\alpha| = j\}$. However, there exists a degree j_* such that for all $j \geq j_*$ the set of observed admissible shifts coincides with a fixed finite set $\Delta_\mathcal{O}$. We refer to this property as *stability* of the delta set.

Assembly at arbitrary degree. Once a stable delta set $\Delta_\mathcal{O}$ has been identified, the matrix representation of $\mathcal{O}$ on Π_n^D for arbitrary n may be constructed as follows:

1. loop over all basis functions (columns) indexed by α,
2. for each α, loop over all $\Delta \in \Delta_\mathcal{O}$,
3. form the candidate target index $\beta = \alpha + \Delta$,
4. if β is admissible, insert the corresponding matrix entry coupling $\phi_\alpha^{(\kappa)}$ to $\phi_\beta^{(\kappa')}$.

The mapping between multi-indices and linear matrix indices is provided by the fixed ordering of the Jacobi basis within each homogeneous subspace and depends on the total degree, but the set of admissible shifts $\Delta_\mathcal{O}$ is independent of n.

Interpretation. In this formulation, the delta set $\Delta_{\mathcal{O}}$ encodes all possible local coupling types induced by the operator in multi-index space. The degree dependence of the matrix sparsity pattern arises solely from the admissibility conditions and the enumeration of basis functions, not from changes in the underlying operator structure. The apparent spreading of nonzeros with increasing degree is a consequence of the growing number of basis functions between two fixed multi-index shifts in the graded ordering; the underlying multi-index coupling encoded by each delta remains unchanged.

Data-driven discovery of the finite delta sets Δ_*

For each operator family $\mathcal{O}$ (e.g. $\mathcal{D}$, $\mathcal{K}$, $\mathcal{J}$) we assume the existence of a finite multi-index shift set

$$\Delta_{\mathcal{O}} \subset \mathbb{Z}^D$$

such that, for every homogeneous degree j and every basis index α with $|\alpha| = j$, the action of $\mathcal{O}$ on $\phi_\alpha^{(\kappa)}$ is supported only on shifts $\alpha + \Delta$ with $\Delta \in \Delta_{\mathcal{O}}$ (cf. the sparsity assumptions above). We describe a practical strategy to discover $\Delta_{\mathcal{O}}$.

Step 0: choose a representative homogeneous block. Fix a test degree n_{test} and select a representative block degree

$$j_{\text{rep}} = j_{\text{rep}}(n_{\text{test}})$$

for extracting the stencil. In our implementation we take

$$j_{\text{rep}} = n_{\text{test}} - 1,$$

i.e. the second highest-degree homogeneous block in $\Pi^D_{n_{\text{test}}}$, which is the largest block and empirically exhibits the full interior coupling.

Step 1: assemble a pruned dense test operator. Assemble a dense matrix representation

$$O^{(n_{\text{test}})} \in \mathbb{R}^{M \times M}, \qquad M = \dim \Pi^D_{n_{\text{test}}},$$

for the operator $\mathcal{O}$ (with the desired parameters κ, shifts, and axis), and apply numerical pruning to remove entries below a relative tolerance (e.g. row-relative threshold proportional to machine epsilon). This produces a matrix whose remaining nonzeros reflect the algebraic sparsity pattern.

Step 2: extract the set of observed deltas from the representative block. Restrict $O^{(n_{\text{test}})}$ to the representative homogeneous block

$$O^{(n_{\text{test}})}_{\text{rep}} = O^{(n_{\text{test}})}\big|_{\mathcal{H}^D_{j_{\text{rep}}} \times \mathcal{H}^D_{j_{\text{rep}}+s}},$$

where the degree shift s is determined by the operator:

$$s = \begin{cases} -1, & \mathcal{O} = \mathcal{D} \\ \in \{-1, 0\}, & \mathcal{O} = \mathcal{K} \\ \in \{-1, 0, +1\}, & \mathcal{O} = \mathcal{J} \text{ (handled per block band)} \end{cases}$$

and where $\mathcal{H}^D_{j_{\text{rep}}+s}$ denotes the appropriate source block. For each nonzero entry

$$\big(O^{(n_{\text{test}})}_{\text{rep}}\big)_{i,j} \neq 0,$$

let α_i and α_j be the (unique) multi-indices associated with the row and column basis functions in the chosen ordering (existence assumed). Define the observed delta

$$\Delta = \alpha_i - \alpha_j \in \mathbb{Z}^D.$$

Collect all such deltas and retain the unique set:

$$\Delta_{\mathcal{O}}^{(n_{\text{test}})} = \left\{ \alpha_i - \alpha_j \,:\, \left(O_{\text{rep}}^{(n_{\text{test}})}\right)_{i,j} \neq 0 \right\}.$$

Step 3: stabilization over increasing n_{test}. Starting from a minimal degree $n_{\min}$ (e.g. $n_{\min} = 5$), repeat Steps 1–2 for $n_{\text{test}} = n_{\min}, n_{\min}+1, \ldots, n_{\max}$, producing a sequence of candidate delta sets $\Delta_{\mathcal{O}}^{(n_{\text{test}})}$. We declare the stencil to have stabilized when

$$\Delta_{\mathcal{O}}^{(n_{\text{test}})} = \Delta_{\mathcal{O}}^{(n_{\text{test}}-1)}$$

(as sets, e.g. after sorting a canonical encoding), and set

$$\Delta_{\mathcal{O}} := \Delta_{\mathcal{O}}^{(n_{\text{test}})}.$$

Usage at arbitrary degree. Once $\Delta_{\mathcal{O}}$ is known, the operator at any target degree n can be assembled by looping over source basis functions (multi-indices α) and applying the admissible shifts:

$$\alpha \mapsto \alpha + \Delta, \qquad \Delta \in \Delta_{\mathcal{O}},$$

discarding invalid targets (e.g. negative components or incorrect total degree), and placing the corresponding nonzero entry in the sparse matrix representation. The conversion between multi-indices and local indices within $\mathcal{H}_j^D$ is performed by fixed, degree-consistent maps induced by the chosen ordering.

7.1 Multiplication operators via lifted operator-valued Clenshaw on the D-simplex

Polynomial spaces. Let $D \geq 1$ and let $\kappa \in \mathbb{R}^{D+1}$ with $\kappa_r > -1/2$. Let $\{\phi_\alpha^{(\kappa)}\}_{|\alpha|\leq n}$ denote the orthonormal Jacobi basis on the simplex Δ^D, ordered in graded lexicographic order. Define

$$\Pi_n^D := \operatorname{span}\{\phi_\alpha^{(\kappa)} : |\alpha| \leq n\}, \qquad M_n := \dim \Pi_n^D.$$

Let

$$\mathcal{H}_j^D := \operatorname{span}\{\phi_\alpha^{(\kappa)} : |\alpha| = j\}, \qquad m_j := \dim \mathcal{H}_j^D = \binom{j+D-1}{D-1}.$$

Then

$$\Pi_n^D = \bigoplus_{j=0}^{n} \mathcal{H}_j^D, \qquad M_n = \sum_{j=0}^{n} m_j.$$

For each j, let $E_j \in \mathbb{R}^{M_n \times m_j}$ denote the canonical embedding of the degree-j block, and E_j^T the corresponding extraction.

Coordinate Jacobi matrices. For each coordinate x_i, $i = 1, \dots, D$, define the Jacobi (multiplication) matrix

$$J_i^{(n)} \in \mathbb{R}^{M_n \times M_n}, \qquad (J_i^{(n)})_{\ell k} = \langle \phi_\ell^{(\kappa)},\, x_i\, \phi_k^{(\kappa)} \rangle_{w_\kappa}.$$

Since the basis is orthonormal and multiplication by x_i is self-adjoint,

$$J_i^{(n)} = (J_i^{(n)})^T.$$

Moreover, multiplication by a coordinate changes total degree by at most one, so

$$E_r^T J_i^{(n)} E_s = 0 \quad \text{unless} \quad r \in \{s-1, s, s+1\}.$$

Define the degree blocks

$$\begin{aligned} A_{i,j} &:= E_j^T J_i^{(p)} E_j \in \mathbb{R}^{m_j \times m_j}, \\ B_{i,j} &:= E_j^T J_i^{(p)} E_{j+1} \in \mathbb{R}^{m_j \times m_{j+1}}, \\ C_{i,j} &:= E_j^T J_i^{(p)} E_{j-1} = B_{i,j-1}^T. \end{aligned}$$

Multiplication as operator functional calculus. Let

$$f(x) = \sum_{|\alpha| \le N} c_\alpha\, \phi_\alpha^{(\kappa)}(x), \qquad q(x) = \sum_{|\beta| \le p} q_\beta\, \phi_\beta^{(\kappa)}(x).$$

Formally,

$$q(x) f(x) = \Phi(x)^T\, q(J_1, \dots, J_D)\, c.$$

To evaluate this stably and without aliasing, we work in a lifted space Π_K^D with

$$K \ge N + p,$$

embed c into Π_K^D, apply the multiplication operator there, and then restrict back to Π_N^D.

Lifted unknowns. Define lifted vectors

$$v_j \in \mathbb{R}^{m_j M_K}, \qquad j = 0, 1, \dots, p,$$

and stack

$$v := \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_p \end{bmatrix} \in \mathbb{R}^{M_p M_K}.$$

Each v_j represents the action of $\Phi_j(J)$ on function-space coefficients in Π_K^D.

Stacked degree recurrence. For each $j \ge 0$, the coordinate recurrences

$$x_i \Phi_j = \Phi_{j-1} C_{i,j} + \Phi_j A_{i,j} + \Phi_{j+1} B_{i,j}, \qquad i = 1, \dots, D,$$

are stacked into

$$\mathcal{C}_j \Phi_{j-1} + \mathcal{A}_j \Phi_j - (I \otimes J^{(K)}) \Phi_j + \mathcal{B}_j \Phi_{j+1} = 0,$$

where

$$\mathcal{A}_j := \begin{bmatrix} A_{1,j} \\ \vdots \\ A_{D,j} \end{bmatrix}, \quad \mathcal{B}_j := \begin{bmatrix} B_{1,j} \\ \vdots \\ B_{D,j} \end{bmatrix}, \quad \mathcal{C}_j := \begin{bmatrix} C_{1,j} \\ \vdots \\ C_{D,j} \end{bmatrix}.$$

This system has Dm_j equations but only m_{j+1} unknowns in Φ_{j+1}.

Canonical pivot rule. Let $\alpha = (\alpha_1, \ldots, \alpha_D)$ denote the graded-lex multi-index. From the Dm_j stacked equations, select exactly m_{j+1} rows by:

- taking all rows from coordinate $i = 1$,
- from coordinate $i = 2$, taking only those basis indices with $\alpha_1 = 0$,
- from coordinate $i = 3$, those with $\alpha_1 = \alpha_2 = 0$,
- $\vdots$
- from coordinate $i = D$, those with $\alpha_1 = \cdots = \alpha_{D-1} = 0$.

The total number of selected rows is

$$m_j + \binom{j+D-2}{D-2} + \cdots + 1 = \binom{j+D}{D-1} = m_{j+1}.$$

Let P_j denote the row-selection operator implementing this pivot rule. Define the effective blocks

$$A_j := P_j \mathcal{A}_j, \qquad B_j := P_j \mathcal{B}_j, \qquad C_j := P_j \mathcal{C}_j.$$

Lifted block system. Define the block operator $L_{p,K}(J) \in \mathbb{R}^{(M_p M_K)\times(M_p M_K)}$ by:

- the $(0,0)$ block is I_{M_K};
- for $j = 0, \ldots, p-1$, the $(j+1)$-st block row is

$$C_j \otimes I_{M_K} \;+\; A_j \otimes I_{M_K} \;-\; (I \otimes J^{(K)}) \;+\; B_j \otimes I_{M_K},$$

  acting on (v_{j-1}, v_j, v_{j+1}).

This operator is square and block lower triangular.

Central identity (operator-valued Clenshaw). Let $q \in \mathbb{R}^{M_p}$ and let $c^{(K)} \in \mathbb{R}^{M_K}$ be the embedding of $c \in \mathbb{R}^{M_N}$ into Π_K^D. Then

$$\boxed{\begin{aligned} &L_{p,K}(J)^T v = q \otimes c^{(K)}, \\ &M_q^{(K)} c^{(K)} = (e_0^T \otimes I)\, v. \end{aligned}}$$

Finally, the de-aliased multiplication operator on Π_N^D is obtained by restriction:

$$\boxed{M_q^{(N)} c = R_{K\to N}\left(M_q^{(K)} E_{N\to K} c\right).}$$

Interpretation. This construction evaluates the multiplication operator $q(J)$ by a lifted, block-triangular solve (operator-valued Clenshaw), avoiding aliasing by performing all intermediate operations in Π_K^D before restriction.

7.1.1 Block back substitution (operator Clenshaw) without assembling $L_{p,K}(J)$

We keep the notation of the previous section. Fix $D \geq 1$, degrees $p \geq 0$ and $K \geq 0$, and let $M_p = \dim \Pi_p^D$, $M_K = \dim \Pi_K^D$, $m_j = \dim \mathcal{H}_j^D$.

Lifted unknowns and right-hand side. For each $j = 0, 1, \dots, p$ define

$$v_j \in \mathbb{R}^{m_j M_K}, \qquad v := [v_0; v_1; \dots; v_p] \in \mathbb{R}^{M_p M_K}.$$

Let the coefficient blocks of $q \in \Pi_p^D$ be $q_j \in \mathbb{R}^{m_j}$ so that $q = [q_0; \dots; q_p] \in \mathbb{R}^{M_p}$. Let $c^{(K)} \in \mathbb{R}^{M_K}$ denote the embedding of $c \in \mathbb{R}^{M_N}$ into Π_K^D (zeros in degrees $> N$). Define the block right-hand side

$$r_j := q_j \otimes c^{(K)} \in \mathbb{R}^{m_j M_K}, \qquad r := [r_0; \dots; r_p] = q \otimes c^{(K)} \in \mathbb{R}^{M_p M_K}.$$

Row selection (canonical pivot rule). For each $j = 0, \dots, p-1$ we stack the D coordinate equations, yielding Dm_j rows. Let P_j be the row-selection operator that picks exactly m_{j+1} rows (the pivot rule used in code):

- from coordinate block $i = 1$: all m_j rows,
- from coordinate block $i = 2$: only those with $\alpha_1 = 0$,
- from coordinate block $i = 3$: only those with $\alpha_1 = \alpha_2 = 0$,
- $\vdots$
- from coordinate block $i = D$: only those with $\alpha_1 = \cdots = \alpha_{D-1} = 0$,

so that the total number of selected rows is m_{j+1}.
Equivalently, there exists a deterministic map

$$\sigma_j : \{1, \dots, m_{j+1}\} \to \{1, \dots, D\} \times \{1, \dots, m_j\}, \qquad \sigma_j(\ell) = (i_\ell, b_\ell),$$

such that the ℓ-th selected row comes from coordinate i_ℓ and within-degree index b_ℓ.

Degree blocks from Jacobi matrices. Using the Jacobi matrices in Π_p^D, define the per-coordinate blocks

$$A_{i,j} := E_j^T J_i^{(p)} E_j \in \mathbb{R}^{m_j \times m_j}, \qquad B_{i,j} := E_j^T J_i^{(p)} E_{j+1} \in \mathbb{R}^{m_j \times m_{j+1}}, \qquad C_{i,j} := B_{i,j-1}^T.$$

Stack across coordinates:

$$\mathcal{A}_j := \begin{bmatrix} A_{1,j} \\ \vdots \\ A_{D,j} \end{bmatrix} \in \mathbb{R}^{(Dm_j) \times m_j}, \quad \mathcal{B}_j := \begin{bmatrix} B_{1,j} \\ \vdots \\ B_{D,j} \end{bmatrix} \in \mathbb{R}^{(Dm_j) \times m_{j+1}}, \quad \mathcal{C}_j := \begin{bmatrix} C_{1,j} \\ \vdots \\ C_{D,j} \end{bmatrix} \in \mathbb{R}^{(Dm_j) \times m_{j-1}}.$$

Apply the pivot rule:

$$A_j := P_j \mathcal{A}_j \in \mathbb{R}^{m_{j+1} \times m_j}, \qquad B_j := P_j \mathcal{B}_j \in \mathbb{R}^{m_{j+1} \times m_{j+1}}, \qquad C_j := P_j \mathcal{C}_j \in \mathbb{R}^{m_{j+1} \times m_{j-1}} \ (j \geq 1).$$

The row-picked function-space multiplication operator. Let $J_i^{(K)} \in \mathbb{R}^{M_K \times M_K}$ be the Jacobi matrices in Π_K^D (the function-space multiplication operators). Define a linear operator

$$\mathcal{J}_j : \mathbb{R}^{m_j M_K} \to \mathbb{R}^{m_{j+1} M_K}$$

by the canonical row map σ_j: write any $u \in \mathbb{R}^{m_j M_K}$ in m_j blocks $u = [u^{(1)}; \ldots; u^{(m_j)}]$ with $u^{(b)} \in \mathbb{R}^{M_K}$, and define

$$(\mathcal{J}_j u)^{(\ell)} := J_{i_\ell}^{(K)}\, u^{(b_\ell)} \in \mathbb{R}^{M_K}, \qquad \ell = 1, \ldots, m_{j+1},$$

where $\sigma_j(\ell) = (i_\ell, b_\ell)$. This is exactly the action assembled in the code by placing the row-picked blocks $-(I \otimes J_i^{(K)})$ into the middle block column of $L_{p,K}(J)$.
The adjoint $\mathcal{J}_j^T : \mathbb{R}^{m_{j+1} M_K} \to \mathbb{R}^{m_j M_K}$ satisfies

$$(\mathcal{J}_j^T w)^{(b)} \;=\; \sum_{\ell:\, b_\ell = b} (J_{i_\ell}^{(K)})^T\, w^{(\ell)} \;=\; \sum_{\ell:\, b_\ell = b} J_{i_\ell}^{(K)}\, w^{(\ell)},$$

using symmetry of $J_i^{(K)}$.

Implicit block operator $L_{p,K}(J)$. Define the block-lower-triangular operator $L_{p,K}(J)$ (never explicitly assembled for the back-substitution form) by its action on block vectors $[v_0; \ldots; v_p]$:

$$\begin{aligned} &\text{block row } 0: && I_{M_K}\, v_0, \\ &\text{block row } (j+1),\ j = 0, \ldots, p-1: && (C_j \otimes I_{M_K})\, v_{j-1} \;+\; \big(A_j \otimes I_{M_K} - \mathcal{J}_j\big)\, v_j \;+\; (B_j \otimes I_{M_K})\, v_{j+1}. \end{aligned}$$

(For $j = 0$, the C_0 term is absent.)

Central solve and extraction. We compute v from the *upper-triangular* system

$$L_{p,K}(J)^T v = r = q \otimes c^{(K)},$$

and return

$$M_q^{(K)} c^{(K)} = (e_0^T \otimes I)\, v = v_0 \in \mathbb{R}^{M_K}.$$

Finally, the de-aliased multiplication operator on Π_N^D is

$$M_q^{(N)} c = R_{K \to N}\, v_0.$$

Block back substitution (operator Clenshaw). Because $L_{p,K}(J)$ is block lower triangular with diagonal blocks

$$D_0 := I_{M_K}, \qquad D_j := B_{j-1} \otimes I_{M_K} \quad (j = 1, \ldots, p),$$

the transposed system $L_{p,K}(J)^T v = r$ is block upper triangular. Introduce the conventions $v_{p+1} \equiv 0$ and $v_{p+2} \equiv 0$. Then the back substitution proceeds for $j = p, p-1, \ldots, 0$ as follows:

- For $j = p$:
$$(B_{p-1}^T \otimes I_{M_K})\, v_p = r_p.$$
- For $1 \le j \le p-1$:
$$(B_{j-1}^T \otimes I_{M_K})\, v_j = r_j - \big(A_j \otimes I_{M_K} - \mathcal{J}_j\big)^T v_{j+1} - (C_{j+1}^T \otimes I_{M_K})\, v_{j+2}.$$
- For $j = 0$:
$$v_0 = r_0 - \big(A_0 \otimes I_{M_K} - \mathcal{J}_0\big)^T v_1 - (C_1^T \otimes I_{M_K})\, v_2.$$

Diagonal solves are small (tensor structure). Since $D_j = B_{j-1} \otimes I_{M_K}$, solving

$$(B_{j-1}^T \otimes I_{M_K}) v_j = s_j$$

reduces to applying B_{j-1}^{-T} on the polynomial-block index while leaving the Π_K^D (function-space) dimension untouched: if $v_j = [v_j^{(1)}; \ldots; v_j^{(m_j)}]$ and $s_j = [s_j^{(1)}; \ldots; s_j^{(m_j)}]$ with blocks in $\mathbb{R}^{M_K}$, then for each fixed function-space coordinate $t \in \{1, \ldots, M_K\}$,

$$\begin{bmatrix} v_j^{(1)}[t] \\ \vdots \\ v_j^{(m_j)}[t] \end{bmatrix} = B_{j-1}^{-T} \begin{bmatrix} s_j^{(1)}[t] \\ \vdots \\ s_j^{(m_j)}[t] \end{bmatrix}.$$

In practice, $B_{j-1} \in \mathbb{R}^{m_j \times m_j}$ is small and dense, so we precompute a dense factorization of B_{j-1}^T and apply it to each of the M_K right-hand sides.

Remarks.

- This recurrence is exactly the block back substitution for the system solved in code by `spsolve(L.T,` $q \otimes c$`)`; it avoids assembling $L_{p,K}(J)$ by applying only the block operators A_j, B_j, C_j and $\mathcal{J}_j, \mathcal{J}_j^T$.
- The only places the large $M_K \times M_K$ Jacobi matrices appear are in the applications of $\mathcal{J}_j$ and $\mathcal{J}_j^T$, which are implemented by sparse SpMV with the $J_i^{(K)}$.
- We actually find that B_{j-1} are sparse - lower triangular and banded with a main diagonal, and only non-zeros below the diagonal in the last few row blocks. Hence, we implement a simple back substitution routine on the sparse upper-triangular system.